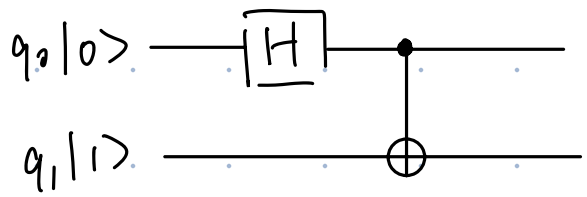
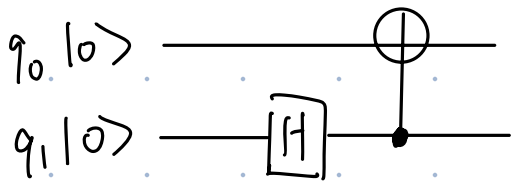
Review CNOT





$$\mathrm{CNOT}_0\,(I\otimes H)(|0\rangle\otimes|0\rangle)$$

$$\mathrm{CNOT}_1\,(H\otimes I)(|0\rangle\otimes|0\rangle)$$

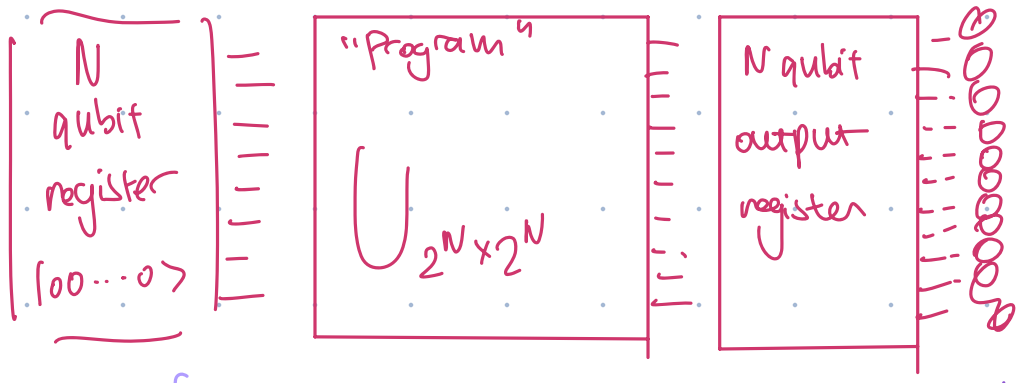
$$= \frac{1}{\sqrt{2}}\left(|00\rangle+|11\rangle\right)$$

$$= \frac{1}{\sqrt{2}}\left(|0\rangle\otimes|0\rangle+|1\rangle\otimes|1\rangle\right)$$

$$\mathrm{CNOT}\left[\frac{1}{\sqrt{2}}\,\underline{|0\rangle}\otimes\underline{\left(|0\rangle+|1\rangle\right)}\right]$$

$$= \mathrm{CNOT}\left[\frac{1}{\sqrt{2}}\left(|00\rangle+|01\rangle\right)\right]$$

$$= \frac{1}{\sqrt{2}}\left[|00\rangle+|11\rangle\right].$$



Theorem: {The CNOT 2-qubit gate + arbitrary single qubit rotations $e^{-i\frac{\theta}{2}\hat{n}\cdot\vec{\sigma}}$} constitute a universal gate set.

↳ can create to some arbitrary precision $U_{2^N\times 2^N}$.

↳ The circuit depth is not exponential.

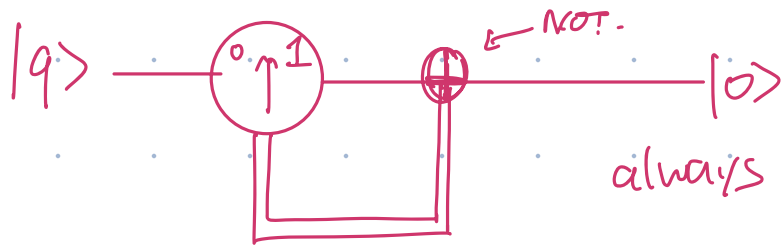
Other resources:

1) Measure
2) Reset.

} both not unitary.

Can I use measurement to create 'reset'?

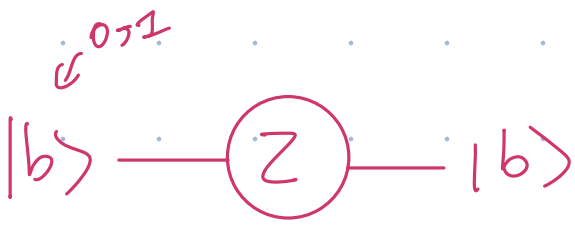
Algorithmic Cooling or reset.



"The Reset" by Measurement.

↳ if 0, do nothing → 0

↳ if 1, flip bit → 0.



$Z^2 = \pm 1$.

$Z = 1 - 2b$

$$B = \frac{1}{2}(I - Z)$$

$$= \frac{1}{2}\left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right]$$

$$= \frac{1}{2}\begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= 0 \cdot |0\rangle\langle 0| + 1 \cdot |1\rangle\langle 1|$$

returns 0 when |0>

& returns 1 when |1>.

$$O = (I \otimes I - Z \otimes Z)/2$$

tells if 2 qubits are in the same state. partial collapse.

$O|00\rangle = 0 \qquad O|01\rangle = +|01\rangle$

$O|11\rangle = 0 \qquad O|10\rangle = +|10\rangle$

$O^2 = O$

$Z \otimes I + I \otimes Z$

complete collapse.

projector

$$O(\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle)$$

$$= \beta|01\rangle + \gamma|10\rangle$$ → this is an entangled state

What if we measure only 1 of 2 qubits?

$$|\psi\rangle = \psi_{00}|\underset{q_1\ q_0}{00}\rangle + \psi_{01}|01\rangle + \psi_{10}|10\rangle + \psi_{11}|11\rangle$$

want to measure q_1 (i.e. want to measure $Z \otimes I$).

do nothing to the q_0

can you measure $B \otimes I$?

$$|\psi\rangle = |0\rangle \otimes [\psi_{00}|0\rangle + \psi_{01}|1\rangle] + |1\rangle \otimes [\psi_{10}|0\rangle + \psi_{11}|1\rangle].$$

↳ can you only do this in a product state?

↳ no.

↳ ??

partial collapse

$Z \otimes I$ possible answers are $+1$ and -1.

if measure $+1$: state collapses to $\dfrac{|0\rangle \otimes [\psi_{00}|0\rangle + \psi_{01}|1\rangle]}{\sqrt{|\psi_{00}|^2 + |\psi_{01}|^2}}$

if measure -1: state collapses to $\dfrac{|1\rangle \otimes [\psi_{10}|0\rangle + \psi_{11}|1\rangle]}{\sqrt{|\psi_{10}|^2 + |\psi_{11}|^2}}$

if a product state: $[\alpha|0\rangle + \beta|1\rangle] \otimes [\gamma|0\rangle + \delta|1\rangle]$

measure $\begin{cases} +1 \\ -1 \end{cases}$

measurement on q_1 doesn't matter on q_0

however, if entangled:

measuring q_1 allows you to know what q_0 is.

$$|\psi\rangle = \sqrt{P_+}\,|0\rangle \otimes \frac{[\psi_{00}|0\rangle + \psi_{01}|1\rangle]}{\sqrt{P_+}} +$$

$$\sqrt{P_-}\,|1\rangle \otimes \frac{[\psi_{10}|0\rangle + \psi_{11}|1\rangle]}{\sqrt{P_-}}$$

$\sqrt{P_+} = \sqrt{|\psi_{00}|^2 + |\psi_{01}|^2}$ $\Rightarrow$ P_+: prob measuring $Z \otimes I \Rightarrow +1$

$\sqrt{P_-} = \sqrt{|\psi_{10}|^2 + |\psi_{11}|^2}$ $\Rightarrow$ P_-: prob measuring $Z \otimes I \Rightarrow -1$

measuring q_1 in an entangled state changes the state of q_0.

$$P_+ = |\psi_{00}|^2 + |\psi_{01}|^2 = P(z_1 = +1) = \sum_{z_0 = \pm 1} P(z_1 = 1, z_0)$$

agrees with classical probability.

$$P(A|B) = \frac{P(A \cup B)}{P(B)}$$

Conditional Probability for measurement result on q_0 given the result on q_0:

$$P(z_0 | z_1) = \frac{P(z_0 \cap z_1)}{P(z_1)}$$ $\rightarrow$ Bayes' Rule.

Born Rule: same w/ classical probability, or different?

$$P(z_1=1, z_0=0) = P(z_1=1)\, P(z_0=1 \mid z_1=1)$$

$$= (\cancel{P_+}) \frac{|\psi_{00}|^2}{\cancel{P_+}}$$

$$= |\psi_{00}|^2$$